

Java Boolean Data Type

The Logic of True and False

1. The Two Choices

A boolean variable is like a digital light switch. It can only ever be in one of two states. In Java, we use lowercase keywords to represent these states.

true

false

```
boolean isJavaFun = true;  
boolean isFishTasty = false;
```

2. Comparison Operators

We often generate booleans by comparing two values. These are called Boolean Expressions.

Operator	Meaning	Example	Result
==	Equal to	10 == 10	true
!=	Not equal	10 != 5	true
>	Greater than	5 > 12	false
<=	Less than or equal	8 <= 8	true

3. Professional Naming

Naming Convention:

Always name your booleans like a **question**. This makes your code read like a sentence.

- isAdult rather than adult
- hasPermission rather than permission
- canDelete rather than delete

4. Practical Use: Decision Making

Booleans are the heart of the `if` statement. The code inside the curly braces only runs if the boolean is **true**.

```
int temperature = 30;
boolean isHot = temperature > 25;

if (isHot) {
    System.out.println("It's a hot day!");
}
```

5. Common Beginner Mistakes

✗ WRONG (String)

```
boolean b = "true";
```

✓ CORRECT

```
boolean b = true;
```

✗ WRONG (Casing)

```
boolean b = True;
```

✓ CORRECT

```
boolean b = true;
```

Summary

Booleans are the foundation of logic. They store **true** or **false**, enable **decision-making**, and represent the **state** of your program.

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